



TnP Connect

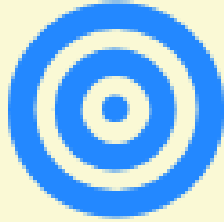
*Unified Training & Placement Management
Solution*

Group Members :

1. Neev Jain (31)
2. Pratyush Pandey (34)
3. Ankush Sahu (45)

CS&E (Cybersecurity) | TCET

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Objectives

TnP Connect - Unified Training & Placement Management Solution

Current Situation	Chaos
WhatsApp Groups	100+ Messages, Multiple group
Email Chains	Multiple fake emails
Notice Boards	Unnoticed or Ignored
Word of Mouth	Missed deadline daily

3 Core Gaps :

- Fragmented Communication
- No Centralization
- No Intelligence
- No Targeting



To create a single platform for the TnP Cell to share all placement-related information in an organized way.

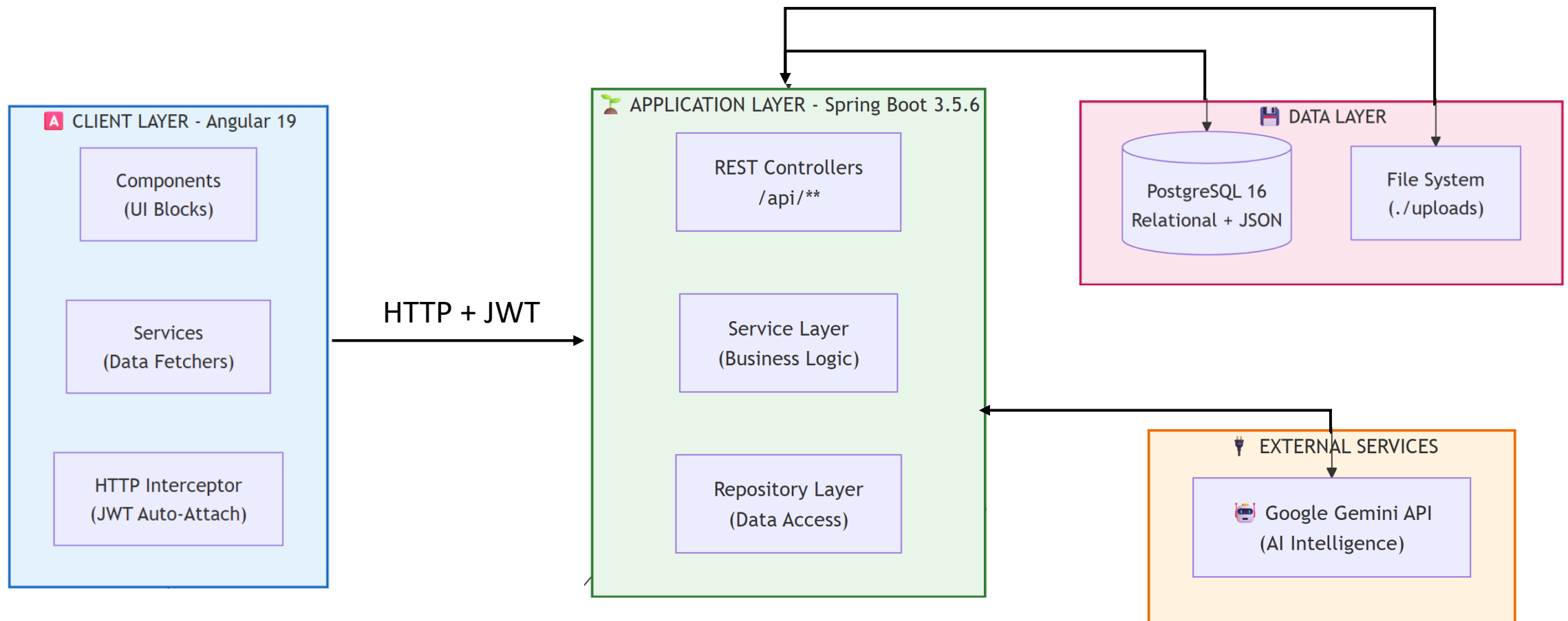


To provide students with easy and quick access to internships, contests, sessions, and important resources.



To improve student engagement and participation in placement activities and events.

Proposed System Architecture



Entity Relationship Model



Student

Central Hub



TnP Admin

Creates/Manages



Internship

1:N with Applications



Session

1:N with Registrations



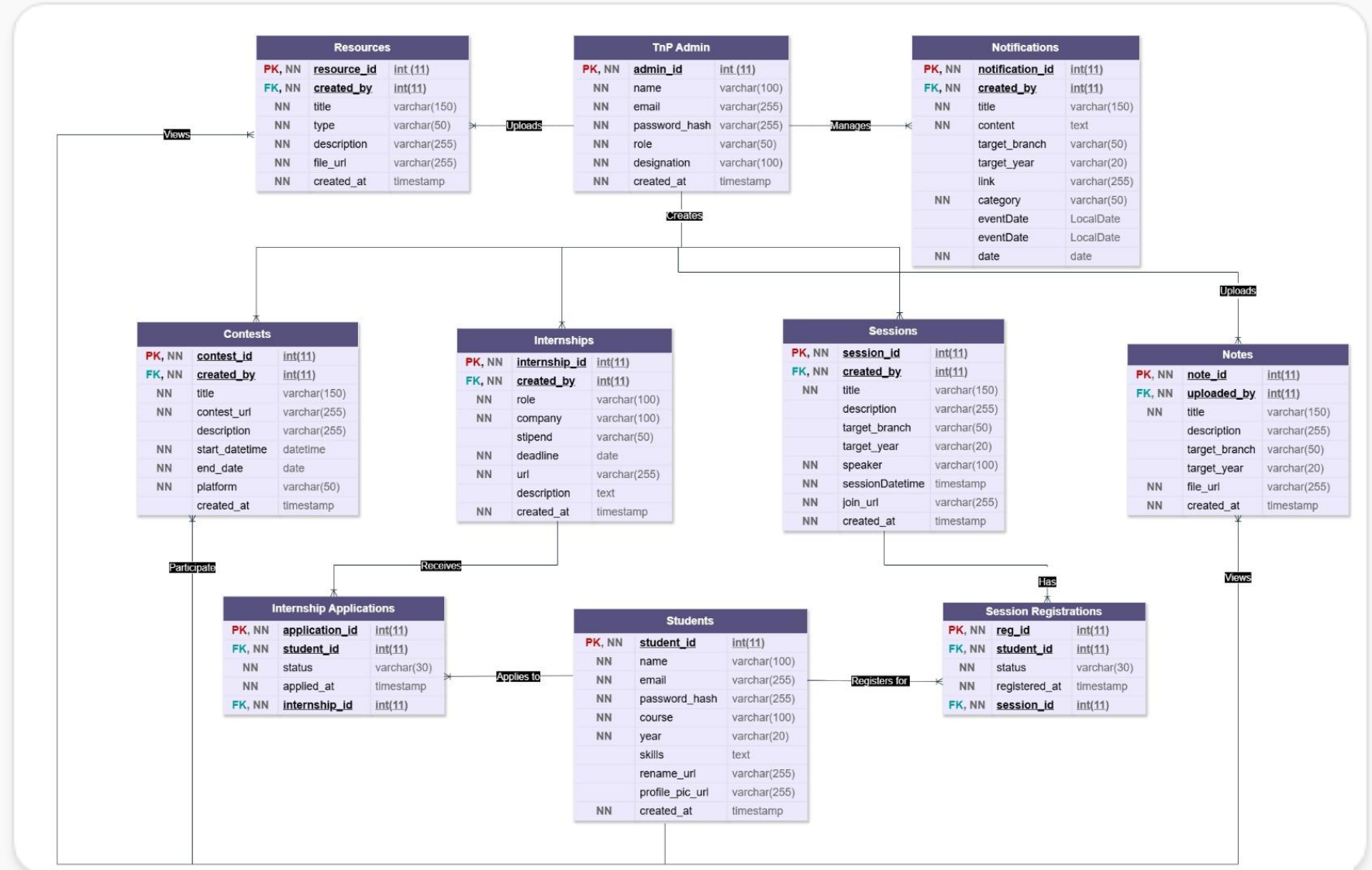
Contest

External Links



Notes/Resources

Branch specific



Modules of TnP Connect

• Student, Admin : Neev Jain, 31

Student

Represents a student user in the system who can view opportunities, register for sessions, apply for internships, and manage their profile details.

Student-controller APIs

GET /api/students/{id}	PATCH /api/students/{id}/change-password
PUT /api/students/{id}	GET /api/students/{id}/skills
DELETE /api/students/{id}	GET /api/students/{id}/session-registrations
PATCH /api/students/{id}	GET /api/students/{id}/internship-applications
GET /api/students/	GET /api/students/{id}/full-details
POST /api/students/	GET /api/students/search

Admin

Represents TnP admins who create, manage, and share internships, sessions, resources, and notifications.

TnP-Admin-controller APIs

POST /api/admins/register
POST /api/admins/login
PATCH /api/admins/{id}
GET /api/admins/{id}/full-details
GET /api/admins/

Modules of TnP Connect

• Internships , Internship Applications : Neev Jain , 31

Internships

Represents internship opportunities in the system, allowing students to view details and apply, while enabling admins to create and manage listings.

Internship-controller APIs

GET /api/internships/

POST /api/internships/

GET /api/internships/{id}

DELETE /api/internships/{id}

GET /api/internships/{id}/applicants

Internship Applications

Tracks which students applied to which internships along with application status and applied date.

Internship-applications-controller APIs

GET /api/applications/

POST /api/applications/

PATCH /api/applications/{id}/status

GET /api/applications/{id}

DELETE /api/applications/{id}

Modules of TnP Connect

• Sessions, Session Registration : Ankush Sahu, 45

Sessions

Represents training sessions with details like title, speaker, date, join link, and target branch/year.

Session-controller APIs

GET /api/sessions/

POST /api/sessions/

GET /api/sessions/{id}

DELETE /api/sessions/{id}

GET /api/sessions/{id}/registrations

Session Registration

Records which students have registered for specific sessions along with their registration status and time.

Session-registration-controller APIs

POST /api/registrations

PATCH /api/registrations/{id}/status

GET /api/registrations/{id}

DELETE /api/registrations/{id}

GET /api/registrations/

Modules of TnP Connect

• Resources, Notes : Ankush Sahu, 45

Resources

Represents learning materials, guides, and files shared by admins for student access.

Resource-controller APIs

GET /api/resources/

POST /api/resources/

GET /api/resources/{id}

DELETE /api/resources/{id}

Notes

Represents notes shared by admins for students, including study or reference materials based on target audience.

Notes-controller APIs

GET /api/notes/

POST /api/notes/

GET /api/notes/{id}

DELETE /api/notes/{id}

Modules of TnP Connect

• Contests : Pratyush Pandey, 34

Contests

Represents contests or hackathons by storing detailed information such as title, description, deadlines, platform, and participation links, allowing students to explore and take part in competitive events.

Contest-controller APIs

GET /api/contests/

POST /api/contests/

GET /api/contests/{id}

DELETE /api/contests/{id}

Modules of TnP Connect

• Notifications : Pratyush Pandey, 34

Notifications

Represents notifications sent by admins to students, containing important updates, events, and announcements targeted based on branch or year.

Notification-controller APIs

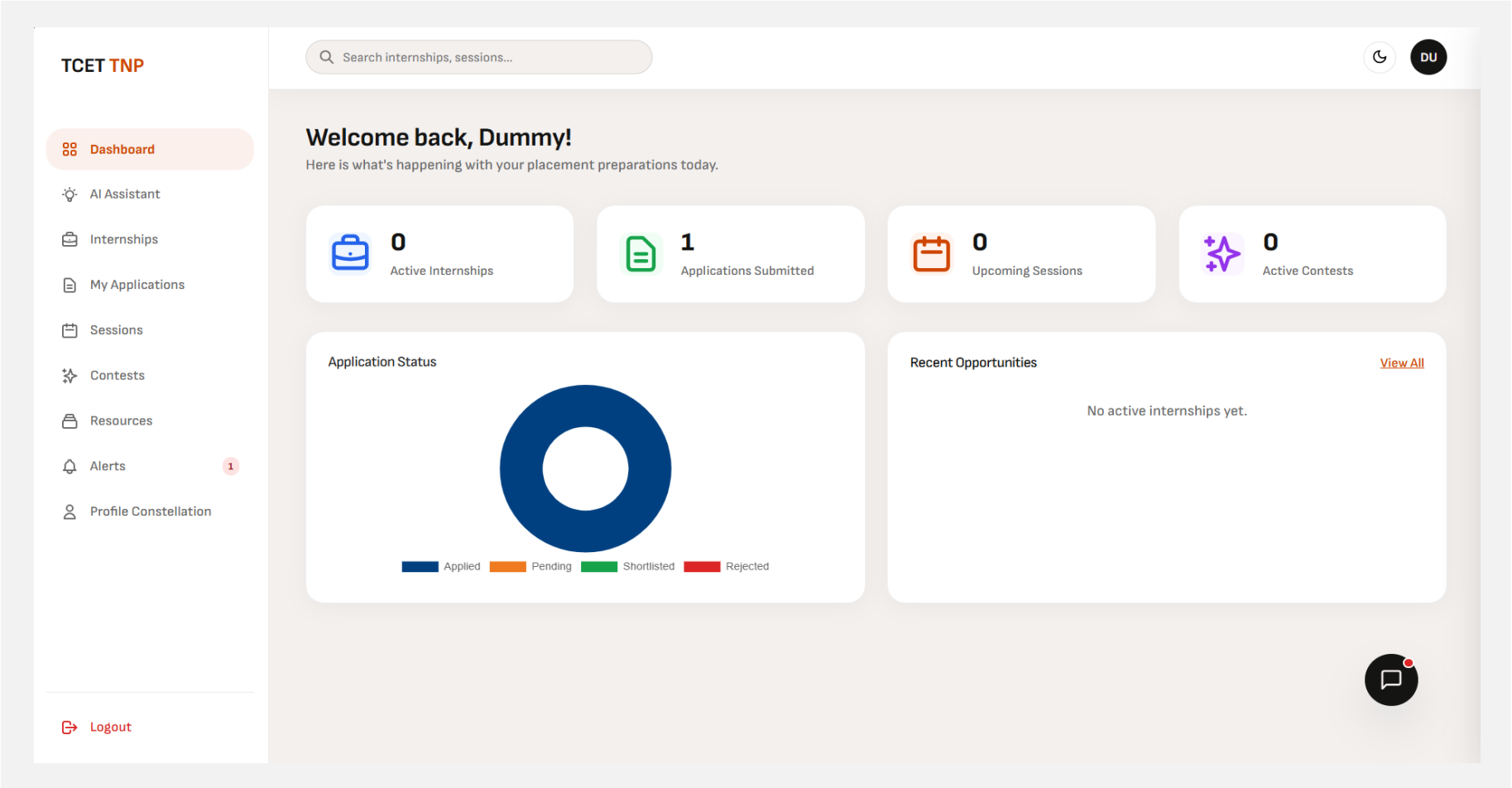
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POST /api/notifications/

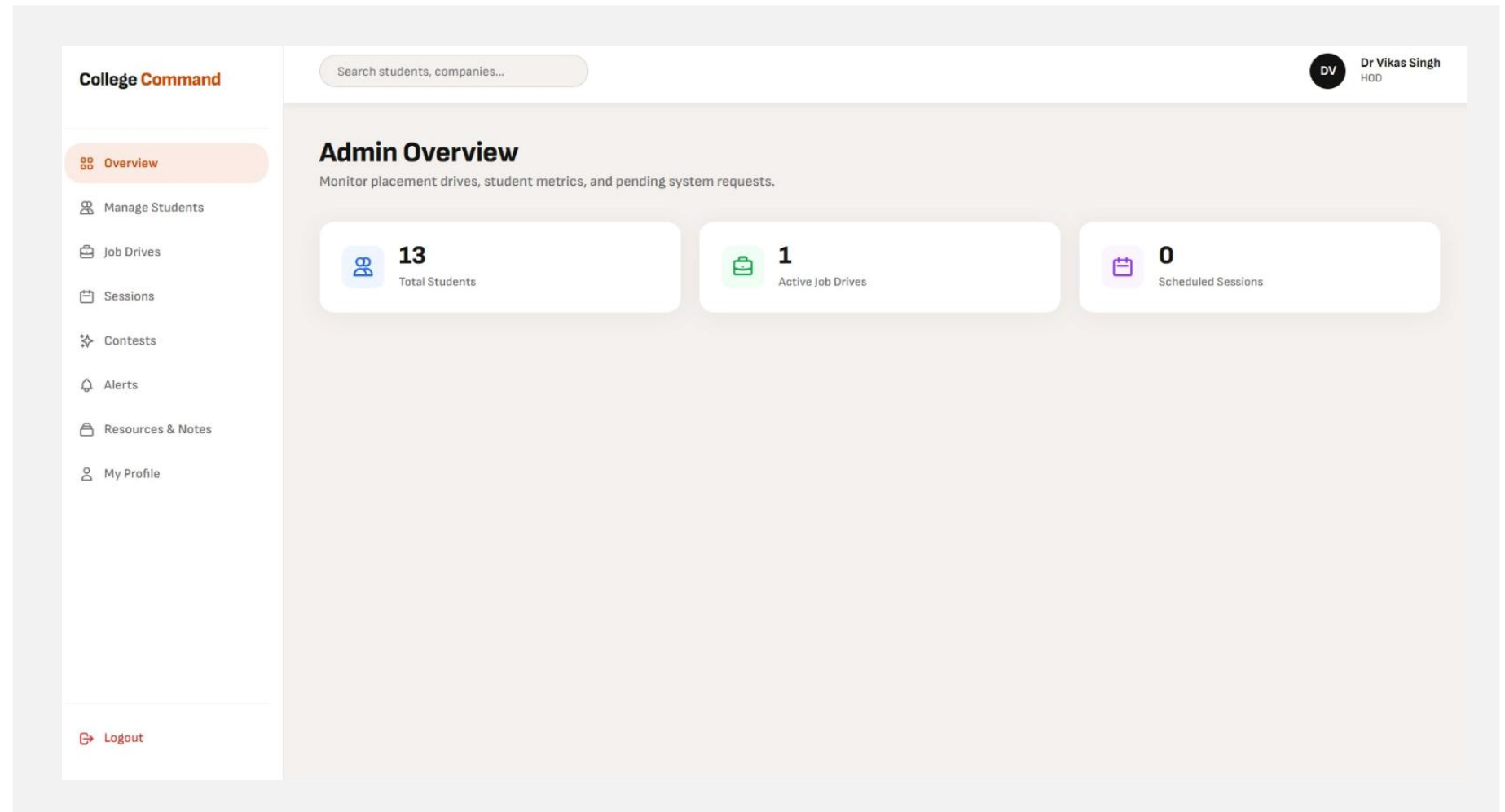
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DELETE /api/notifications/{id}

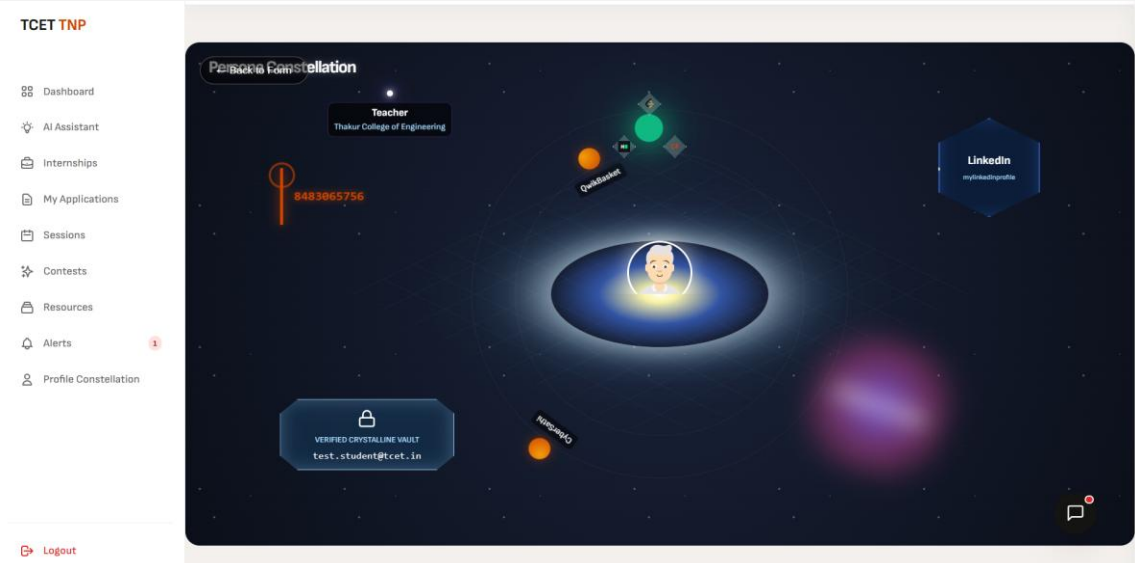
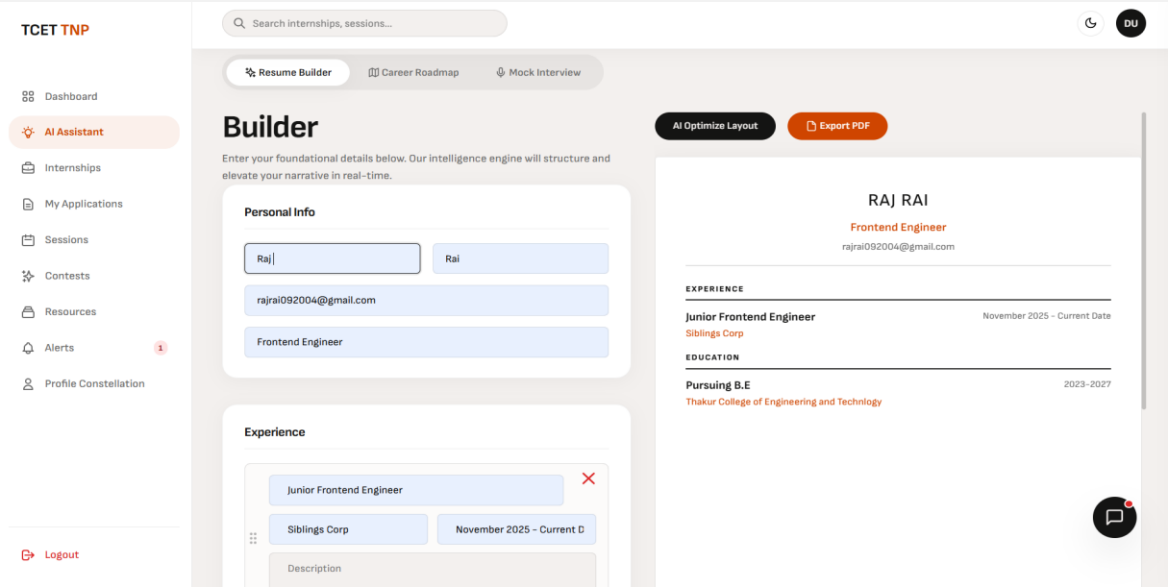
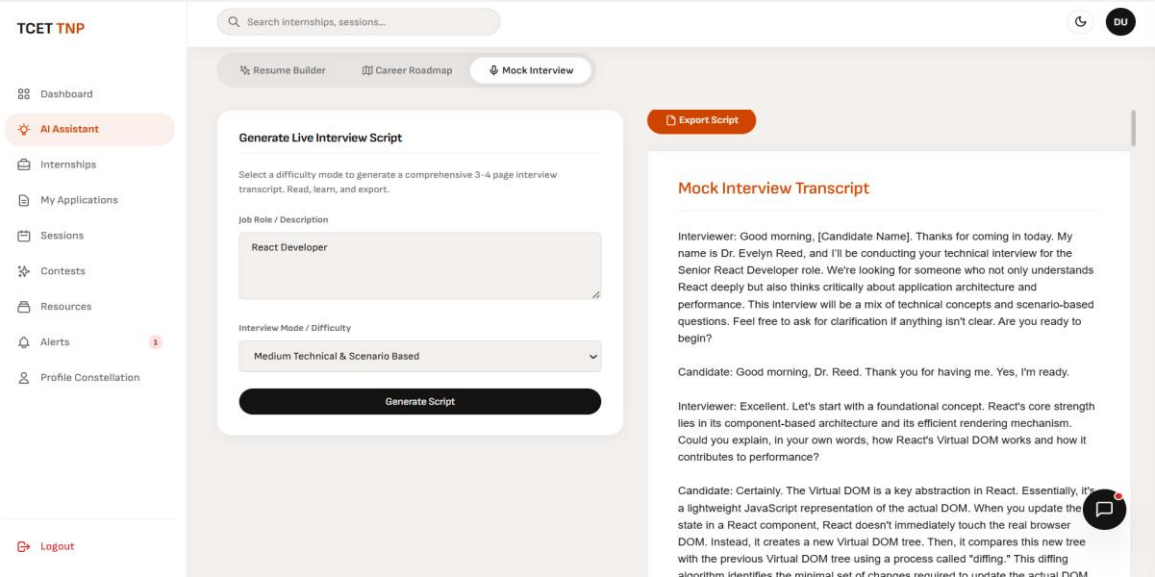
Student
Dashboard



Admin Dashboard



AI Features



Core Implementation Highlights



JWT Authentication

Implementation:

Utilized JSON Web Tokens (JWT) to implement Role-Based Access Control (RBAC).

Project Specifics:

- Securely separates the Placement Head Dashboard (admin controls) from the Student Portal (viewing/applying).
- Tokens are automatically attached to API requests (using HTTP Interceptors in Angular), allowing users to browse internships, use the AI assistant, or apply for roles without needing to repeatedly log in.
- Ensures state management so that if a student refreshes the page, their session remains active until the token expires.

Core Implementation Highlights



Crud Operations

Implementation:

Developed complete RESTful API integration to handle Create, Read, Update, and Delete operations seamlessly between the frontend and database.

Project Specifics:

- **For the Placement Head:** Enables dynamic posting, editing, and removal of core system entities including Internships, Hackathons/Contests, Workshops, and targeted Notifications.
- **For the Students:** Allows students to build and update their digital profiles (adding skills, education, target roles) and track their real-time application statuses.
- Ensures the live feed on the student dashboard updates immediately when the Placement cell posts a new opportunity.

Core Implementation Highlights



Validators

Implementation:

Integrated strict client-side and server-side validation to ensure clean data enters the database.

Project Specifics:

- **Form Controls:** Applied to critical user inputs such as Student Registration, Login authentication, and Profile building.
- **Custom Rules:** Prevents incomplete applications or incorrect formats (e.g., ensuring valid email structures, mandating specific fields like "Roll No." or "Branch", and verifying password strength).
- Improves user experience by providing immediate, real-time visual feedback (error messages or red highlights) before the user even clicks "Submit."

Core Implementation Highlights



Responsive UI

Implementation:

Built a dynamic, responsive frontend using Angular and custom CSS, seamlessly integrated with a robust Spring Boot backend API.

Project Specifics:

- **Custom Design:** A modern, mobile-friendly interface that adapts perfectly to any screen size for both students and admins.
- **Smart Dynamic Filters:** Collapsible sidebars let users instantly sort internships and workshops without reloading the page.
- **Integrated AI Tools:** Features a floating chat assistant for quick guidance and an AI Resume Generator for instant PDF creation.

THANK YOU

Questions & Feedback Welcome

TnP_Connect: Bridging the Gap Between Academia and Industry

Link: <https://tnp-connect.netlify.app/auth/login>

